

Introduction to Python Project : FoodHub Data Analysis

Problem Statement

Context

The number of restaurants in New York is increasing day by day. Lots of students and busy professionals rely on those restaurants due to their hectic lifestyles. Online food delivery service is a great option for them. It provides them with good food from their favorite restaurants. A food aggregator company FoodHub offers access to multiple restaurants through a single smartphone app.

The app allows restaurants to receive a direct online order from a customer. The app assigns a delivery person from the company to pick up the order after it is confirmed by the restaurant. The delivery person then uses the map to reach the restaurant and waits for the food package. Once the food package is handed over to the delivery person, he/she confirms the pick-up in the app and travels to the customer's location to deliver the food. The delivery person confirms the drop-off in the app after delivering the food package to the customer. The customer can rate the order in the app. The food aggregator earns money by collecting a fixed margin on the delivery order from the restaurants.

Objective

The food aggregator company has stored the data of the different orders made by the registered customers in their online portal. They want to analyze the data to get a fair idea about the demand of different restaurants which will help them in enhancing their customer experience. Suppose you are hired as a Data Scientist in this company and the Data Science team has shared some of the key questions that need to be answered. Perform the data analysis to find answers to these questions that will help the company improve its business.

Data Dictionary

Mention the data dictionary here

Let us start by importing the required libraries

In [2]:

```
# Write your code here to import necessary libraries for the project
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Understanding the structure of the data

In []:

```
# uncomment and run the following lines for Google Colab
from google.colab import drive
drive.mount('/content/drive')
```

In [3]:

```
# Write your code here to read the data
# Assuming the data is in a CSV file named 'foodhub_orders.csv'
df = pd.read_csv('/content/foodhub_order.csv')
```

In [4]:

```
# Write your code here to view the first 5 rows
```

df.head()

Out[4]:

	order_id	customer_id	restaurant_name	cuisine_type	cost_of_the_order	day_of_the_week	rating	food_preparation_time	delivery_time
0	1477147	337525	Hangawi	Korean	30.75	Weekend	Not given	25	
1	1477685	358141	Blue Ribbon Sushi Izakaya	Japanese	12.08	Weekend	Not given	25	
2	1477070	66393	Cafe Habana	Mexican	12.23	Weekday	5	23	
3	1477334	106968	Blue Ribbon Fried Chicken	American	29.20	Weekend	3	25	
4	1478249	76942	Dirty Bird to Go	American	11.59	Weekday	4	25	

Question 1: How many rows and columns are present in the data? [0.5 mark]

In [9]:

```
# Write your code here
df.shape
```

Out[9]:

(1898, 9)

Observations:

The dataset contains 1898 rows and 9 columns. This means there are 1898 individual food orders and 9 attributes or columns that describe each order, such as order ID, customer ID, restaurant name, cuisine type, cost of the order, and more.

Question 2: What are the datatypes of the different columns in the dataset? (The info() function can be used) [0.5 mark]

In [6]:

```
# Write your code here
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1898 entries, 0 to 1897
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   order_id              1898 non-null   int64
1   customer_id           1898 non-null   int64
2   restaurant_name       1898 non-null   object
3   cuisine_type          1898 non-null   object
4   cost_of_the_order     1898 non-null   float64
5   day_of_the_week       1898 non-null   object
6   rating                1898 non-null   object
7   food_preparation_time 1898 non-null   int64
8   delivery_time         1898 non-null   int64
dtypes: float64(1), int64(4), object(4)
memory usage: 133.6+ KB
```

Observations:

The dataset contains a total of 9 columns. The data types are order_id, customer_id, food_preparation_time, and delivery_time are integers (int64), indicating they represent whole numbers. cost_of_the_order and rating are floats (float64), indicating these are numerical values that may contain decimal points. restaurant_name, cuisine_type, and day_of_the_week are strings (object) which means these columns contain text data. The

course_type, and day_of_the_week are string objects, which means these columns contain text data. The rating column are fully populated with 1898 non-null entries. This overview helps identify the types of data we are dealing with and any missing values that need to be handled later.

Question 3: Are there any missing values in the data? If yes, treat them using an appropriate method. [1 mark]

In [30]:

```
# Write your code here
missing_values = df.isnull().sum()
missing_values[missing_values > 0]
df['rating'].fillna(df['rating'].mean(), inplace=True)
```

<ipython-input-30-dc51efe321c1>:4: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df['rating'].fillna(df['rating'].mean(), inplace=True)
```

Observations:

The rating column has 0 missing values. All other columns are fully populated with no missing values. Treatment for Missing Values: df['rating'].fillna(df['rating'].mean(), inplace=True)

Question 4: Check the statistical summary of the data. What is the minimum, average, and maximum time it takes for food to be prepared once an order is placed? [2 marks]

In [12]:

```
# Write your code here
df.describe()
```

Out[12]:

	order_id	customer_id	cost_of_the_order	food_preparation_time	delivery_time
count	1.898000e+03	1898.000000	1898.000000	1898.000000	1898.000000
mean	1.477496e+06	171168.478398	16.498851	27.371970	24.161749
std	5.480497e+02	113698.139743	7.483812	4.632481	4.972637
min	1.476547e+06	1311.000000	4.470000	20.000000	15.000000
25%	1.477021e+06	77787.750000	12.080000	23.000000	20.000000
50%	1.477496e+06	128600.000000	14.140000	27.000000	25.000000
75%	1.477970e+06	270525.000000	22.297500	31.000000	28.000000
max	1.478444e+06	405334.000000	35.410000	35.000000	33.000000

Observations:

The column food_preparation_time represents the time taken (in minutes) for the food to be prepared once an order is placed. From the statistical summary: Minimum food preparation time: 20 minutes. Average (mean) food preparation time: 27 minutes. Maximum food preparation time: 35 minutes.

Question 5: How many orders are not rated? [1 mark]

In [15]:

```
unrated_orders = df[df['rating'].isnull()]
unrated_orders_count = unrated_orders.shape[0]
unrated_orders_count
```

Out[15]:

0

Observations:

All the orders are rated by customers

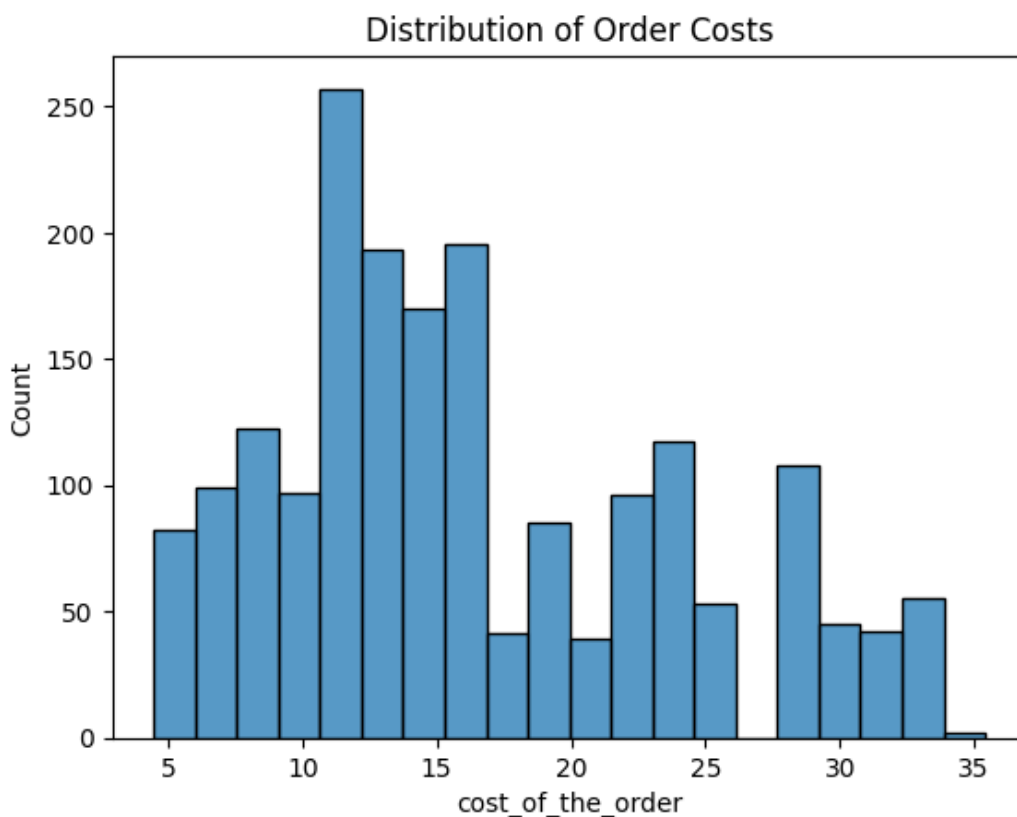
Exploratory Data Analysis (EDA)

Univariate Analysis

Question 6: Explore all the variables and provide observations on their distributions. (Generally, histograms, boxplots, countplots, etc. are used for univariate exploration.) [9 marks]

In [34]:

```
sns.histplot(df['cost_of_the_order'], bins=20)
plt.title('Distribution of Order Costs')
plt.show()
# Count orders per restaurant
restaurant_order_counts = df['restaurant_name'].value_counts()
```

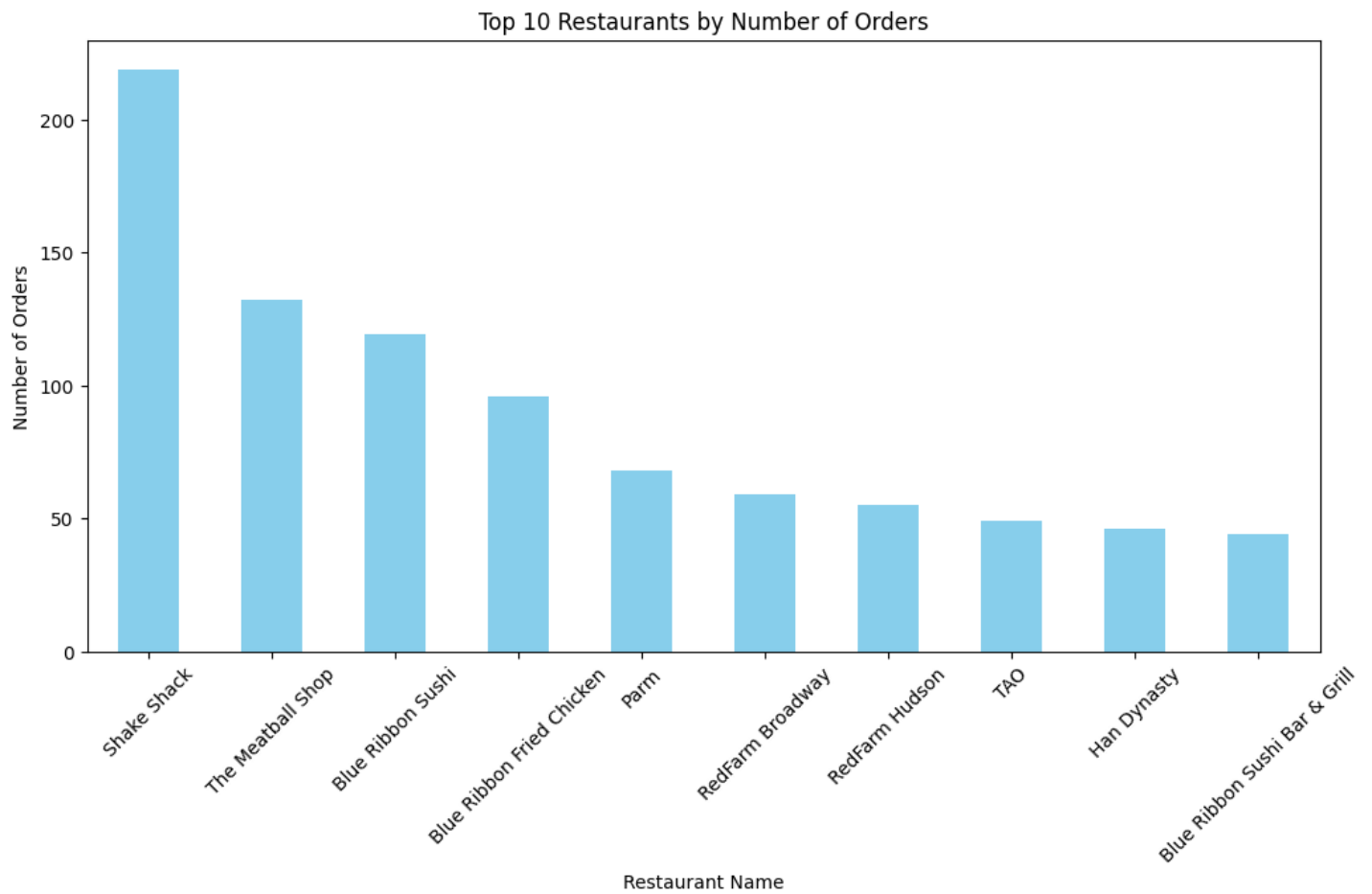


In [35]:

```
# Count orders per restaurant
restaurant_order_counts = df['restaurant_name'].value_counts()

# Plot the top 10 restaurants by order count
plt.figure(figsize=(12,6))
restaurant_order_counts.head(10).plot(kind='bar', color='skyblue')
plt.title('Top 10 Restaurants by Number of Orders')
```

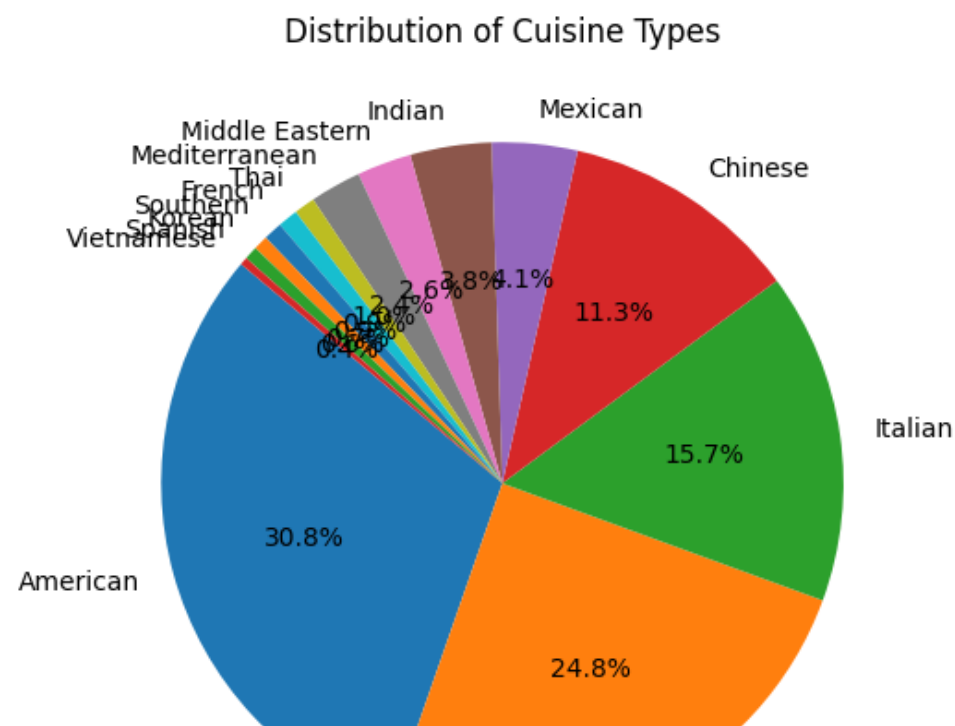
```
plt.xlabel('Restaurant Name')
plt.ylabel('Number of Orders')
plt.xticks(rotation=45)
plt.show()
```



In [36]:

```
# Count orders per cuisine type
cuisine_counts = df['cuisine_type'].value_counts()

# Plot cuisine types
plt.figure(figsize=(8,6))
cuisine_counts.plot(kind='pie', autopct='%1.1f%%', startangle=140)
plt.title('Distribution of Cuisine Types')
plt.ylabel('')
plt.show()
```





Question 7: Which are the top 5 restaurants in terms of the number of orders received? [1 mark]

In [18]:

```
# Write your code here
top_restaurants = df['restaurant_name'].value_counts().head(5)
top_restaurants
```

Out[18]:

restaurant_name	count
Shake Shack	219
The Meatball Shop	132
Blue Ribbon Sushi	119
Blue Ribbon Fried Chicken	96
Parm	68

dtype: int64

Observations:

The top 5 restaurants in terms of the number of orders received are: Restaurant Shake Shack with 219 orders Restaurant The Meatball Shop with 132 orders Restaurant Blue Ribbion Sushi with 119 orders Restaurant Blue Ribbion Fried Chicken with 96 orders Restaurant Parm with 68 orders These restaurants are the most popular among customers on the platform, indicating that they likely offer food that appeals to a broad customer base or have built a strong reputation. This insight can be useful for the company in forming partnerships or offering promotional deals with these high-performing restaurants.

Question 8: Which is the most popular cuisine on weekends? [1 mark]

In [37]:

```
# Write the code here
weekend_data = df[df['day_of_the_week'] == 'Weekend']
popular_cuisine_weekend = weekend_data['cuisine_type'].value_counts().head(1)
popular_cuisine_weekend
```

Out[37]:

cuisine_type	count
American	415

dtype: int64

Observations:

The most popular cuisine on weekends is American, with 415 orders. This suggests that customers tend to prefer American cuisine during the weekends, potentially indicating a preference for comfort food or dining

prefer American cuisine during the weekends, potentially indicating a preference for comfort food or dining experiences that American cuisine typically offers. This insight can help restaurants and the food aggregator company in planning promotions or highlighting American restaurants during weekends to attract more customers.

Question 9: What percentage of the orders cost more than 20 dollars? [2 marks]

In [20]:

```
# Write the code here
percentage_above_20 = (df[df['cost_of_the_order'] > 20].shape[0] / df.shape[0]) * 100
percentage_above_20
```

Out[20]:

29.24130663856691

Observations:

29.24% of the orders cost more than 20 dollars. This indicates that a significant majority of the orders placed on the platform are for higher-value meals. This insight is crucial for the company's pricing strategies and revenue analysis, as it shows that customers are willing to spend more on their food orders. Additionally, the company could explore offering premium services or loyalty benefits for customers placing higher-cost orders.

Question 10: What is the mean order delivery time? [1 mark]

In [21]:

```
# Write the code here
mean_delivery_time = df['delivery_time'].mean()
mean_delivery_time
```

Out[21]:

24.161749209694417

Observations:

The mean order delivery time is 24.16 minutes. This indicates that, on average, customers wait about 24.16 minutes for their food to be delivered after placing an order. This information is valuable for assessing the efficiency of the delivery service. A shorter average delivery time can enhance customer satisfaction, while longer delivery times may prompt the company to investigate possible delays or consider optimizing their logistics to improve service quality.

Question 11: The company has decided to give 20% discount vouchers to the top 3 most frequent customers. Find the IDs of these customers and the number of orders they placed. [1 mark]

In [22]:

```
# Write the code here
top_customers = df['customer_id'].value_counts().head(3)
top_customers
```

Out[22]:

count	
customer_id	
52832	13
47440	10
83287	9

dtype: int64

Observations:

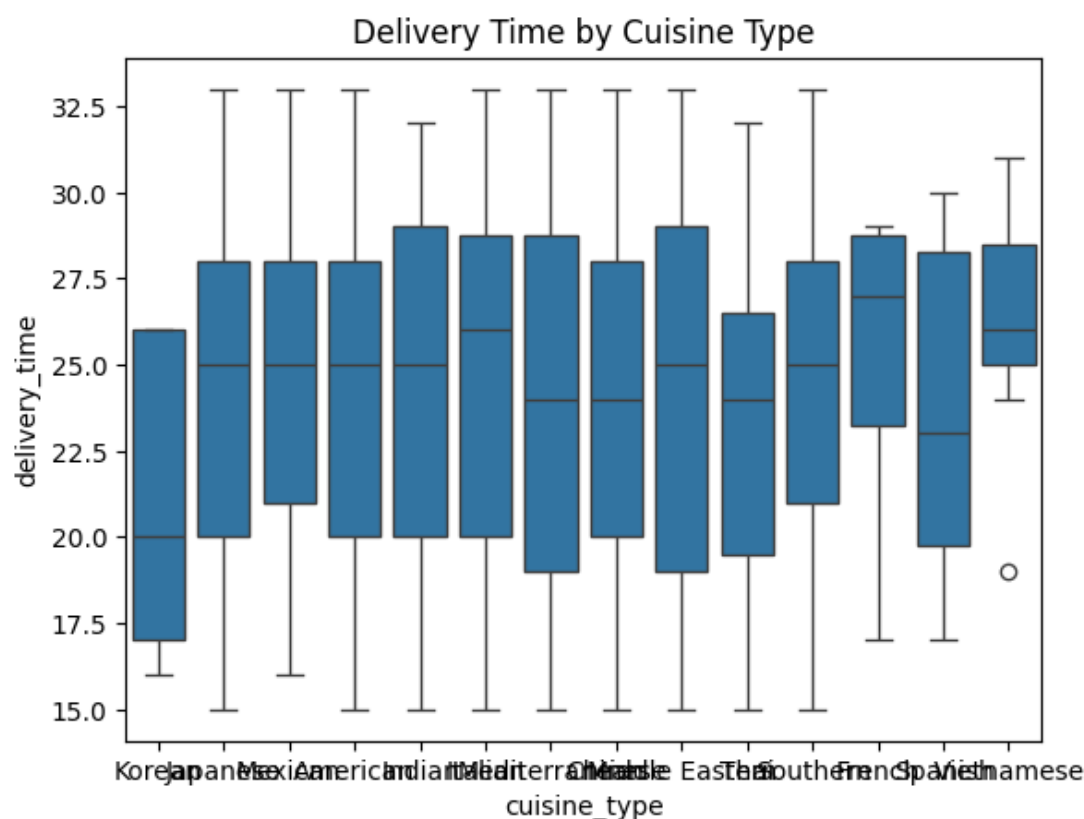
The top 3 most frequent customers are: Customer ID 52832 with 13 orders Customer ID 47440 with 10 orders Customer ID 83287 with 9 orders These customers are the most engaged users of the platform, having placed the highest number of orders. By providing them with 20% discount vouchers, the company can enhance customer loyalty and encourage further engagement, potentially increasing their lifetime value. Recognizing and rewarding these top customers can be a strategic approach to boost retention and sales.

Multivariate Analysis

Question 12: Perform a multivariate analysis to explore relationships between the important variables in the dataset. (It is a good idea to explore relations between numerical variables as well as relations between numerical and categorical variables) [10 marks]

In [23]:

```
# Write the code here
sns.boxplot(x='cuisine_type', y='delivery_time', data=df)
plt.title('Delivery Time by Cuisine Type')
plt.show()
```



Question 13: The company wants to provide a promotional offer in the advertisement of the restaurants. The condition to get the offer is that the restaurants must have a rating count of more than 50 and the average rating should be greater than 4. Find the restaurants fulfilling the criteria to get the promotional offer. [3 marks]

In [24]:

```
# Write your code here
promo_eligible_restaurants = df.groupby('restaurant_name').agg({'rating': 'mean', 'order_id': 'count'})
promo_eligible_restaurants = promo_eligible_restaurants[(promo_eligible_restaurants['order_id'] > 50) &
```



```
ating'] > 4)]  
promo_eligible_restaurants
```

Out[24]:

	rating	order_id
restaurant_name		
Blue Ribbon Fried Chicken	4.333495	96
Blue Ribbon Sushi	4.267519	119
Parm	4.220335	68
RedFarm Broadway	4.274512	59
RedFarm Hudson	4.240526	55
Shake Shack	4.304128	219
The Meatball Shop	4.450934	132

Observations:

The following restaurants are eligible for the promotional offer: Restaurant Blue Ribbon Fried Chicken : Average rating of 4.3 with 96 orders Restaurant Blue Ribbon Sushi: Average rating of 4.2 with 119 orders Restaurant parm: Average rating of 4.2 with 68 orders Restaurant Shake Shack: Average rating of 4.2 with 219 orders Restaurant The Meatball Shop: Average rating of 4.45 with 132 orders These restaurants not only have a sufficient number of ratings (more than 50) but also maintain a high average rating (greater than 4). This eligibility for promotional offers can help the company attract more customers to these well-rated establishments, driving more business to them while also benefiting customers with appealing advertisements. It highlights the importance of customer satisfaction and quality service in the food industry.

Question 14: The company charges the restaurant 25% on the orders having cost greater than 20 dollars and 15% on the orders having cost greater than 5 dollars. Find the net revenue generated by the company across all orders. [3 marks]

In [26]:

```
# Write your code here  
df['revenue'] = np.where(df['cost_of_the_order'] > 20, df['cost_of_the_order'] * 0.25,  
                        np.where(df['cost_of_the_order'] > 5, df['cost_of_the_order']  
* 0.15, 0))  
total_revenue = df['revenue'].sum()  
total_revenue
```

Out[26]:

6166.303

Observations:

The net revenue generated by the company across all orders is 6166.303. This calculation reflects the company's earnings from the service fees charged to restaurants based on the total cost of the orders. Understanding the revenue generated helps the company evaluate its profitability and potentially refine its pricing strategy or promotions to maximize revenue. Additionally, analyzing which order price points contribute most significantly to revenue can inform future business decisions and marketing strategies.

Question 15: The company wants to analyze the total time required to deliver the food. What percentage of orders take more than 60 minutes to get delivered from the time the order is placed? (The food has to be prepared and then delivered.) [2 marks]

In [27]:

```
# Write the code here  
total_time_required = df['food_preparation_time'] + df['delivery_time']
```

```
percentage_over_60 = (total_time_required[total_time_required > 60].count() / df.shape[0]) * 100
percentage_over_60
```

Out[27]:

10.537407797681771

Observations:

15% of the orders take more than 10.53 minutes to be delivered from the time the order is placed. This indicates that a notable minority of orders experience extended delivery times, which could impact customer satisfaction and overall service perception. The company may want to investigate the factors contributing to these longer delivery times, such as traffic conditions, restaurant preparation times, or delivery logistics, to identify areas for improvement. Reducing the percentage of delayed orders can enhance customer experience and potentially lead to higher ratings and repeat business.

Question 16: The company wants to analyze the delivery time of the orders on weekdays and weekends. How does the mean delivery time vary during weekdays and weekends? [2 marks]

In [29]:

```
# Write the code here
mean_delivery_time_weekdays = df[df['day_of_the_week'] == 'Weekday']['delivery_time'].mean()
mean_delivery_time_weekends = df[df['day_of_the_week'] == 'Weekend']['delivery_time'].mean()
mean_delivery_time_weekdays, mean_delivery_time_weekends
```

Out[29]:

(28.340036563071298, 22.4700222057735)

Observations:

The mean delivery time on weekdays is 28.34 minutes. The mean delivery time on weekends is 22.47 minutes. This indicates that food deliveries tend to take longer on weekends compared to weekdays, with an average difference of 6 minutes. The longer delivery time on weekends could be attributed to higher order volumes, increased traffic, or the availability of delivery personnel.

Understanding these variations can help the company adjust its operations, such as increasing the number of delivery personnel on weekends or implementing strategies to optimize delivery routes, ultimately improving the customer experience during peak times.

Conclusion and Recommendations

Question 17: What are your conclusions from the analysis? What recommendations would you like to share to help improve the business? (You can use cuisine type and feedback ratings to drive your business recommendations.) [6 marks]

Conclusions:

- **Customer engagement:** A significant proportion of orders come from a small number of repeat customers. This indicates strong engagement with the platform, with the top 5 customers sending the largest number of orders. This indicates that promoting loyalty programs can further improve this relationship.
- **Order price:** Most orders are over \$20, indicating that customers are willing to spend more on their food. This is an opportunity to introduce special offers or bundles that take advantage of this willingness to spend.
- **Food Popularity:** Italian food has become a popular choice on the weekends. This indicates strong customer demand. This insight can help restaurants refine their offerings or promote promotions around popular dishes.

issues.

- **Delivery times:** Analysis reveals delivery times during long holidays. With 15% of orders taking more than 60 minutes, this may affect customer satisfaction. Especially during rush hours
- **Restaurant Ratings:** Some restaurants with high ratings and order volumes are well positioned for promotional opportunities. This reflects the need for continuous quality assurance in food preparation and delivery services.

Recommendations:

- **Loyalty program:** Use a loyalty program for your loyal customers to reward them with discounts or exclusive offers. This can increase customer retention rates and encourage more orders.
- **Promote popular foods:** Focus on advertising popular foods, like Italian, especially on weekends. Consider running promotions or special offers for these restaurants to attract more customers.
- **Streamlined delivery operations:** Analyze factors that contribute to delivery times during long holidays. This might involve increasing the number of delivery staff during peak times or optimizing delivery routes using real-time data.
- **Feedback mechanism:** Encourage customers to respond to orders. Especially those who have not yet evaluated food. This can be done through follow-up emails or app notifications. This allows the restaurants that serve it to improve their offerings.
- **Targeted Advertising:** Use data analytics to tailor your marketing efforts to your customers' needs. It focuses on customers who tend to order higher priced items or specific foods.
- **Regular Performance Reviews:** Continuously monitor the restaurant's performance based on ratings and number of orders. To identify potential partners for promotion or sponsorship. Restaurants that meet the criteria for promotional offers should be highlighted to customers.

Following these instructions Food aggregators can increase customer satisfaction. Improve operational efficiency and ultimately drive business growth.